

Shreyas M Kaushik

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SUMMARY

Senior Software Engineer with 4+ years building AI/ML systems, agentic platforms, and cloud-native infrastructure at scale. Currently at Mphasis.AI, shipping an on-premises agentic platform and an enterprise LLM gateway for BFSI clients. Strong expertise in Python, TypeScript, infrastructure, and AI-driven development with a habit of owning systems end-to-end.

TECHNICAL SKILLS

- **Languages:** TypeScript, Python, Golang, JavaScript, Bash, SQL
- **Web Technologies:** Node.js, React.js, Django, FastAPI, Express, RESTful APIs, NGINX
- **Agentic AI & LLMs:** LLM Gateway design, Multi-agent orchestration, Tool use, RAG pipelines, OpenAI/Anthropic/Gemini APIs
- **Workflow Orchestration:** Netflix Conductor, Temporal, Event-driven architecture
- **AI/ML Integration:** Model deployment, Scalability, Classification pipelines
- **Cloud & DevOps:** AWS (EC2, S3, SQS, ALB, EKS, API Gateway, Cognito), Docker, Kubernetes, Helm, GitHub Actions
- **Databases:** MongoDB, PostgreSQL, Redis, Redis Sentinel, TypeORM
- **Monitoring & Security:** Grafana, Loki, Tempo, Promtail, Prometheus, OPA, Keycloak, OAuth 2.0, JWT, MFA

EXPERIENCE

- **Mphasis** Bangalore, India
Senior Software Engineer, Applied AI 16-02-2026 - Present
 - **ATOM - Enterprise Edition:** *AgentScope, TypeScript, Golang, FastAPI, Docker, Kubernetes, Kafka (Redpanda), OPA, PostgreSQL, Redis, Grafana Alloy, Loki, Tempo*
Extended ATOM (my open-source agentic platform) into an enterprise-grade product for BFSI clients, adding production hardening, BFSI-specific OPA compliance policies, PCI-DSS/GDPR/DORA guardrails, and multi-tenant isolation. The platform lets teams write agents via SDK and deploy with a single command - every LLM call flows through GATE (JWT auth, OPA policy, rate limits, immutable audit log) with agents running as isolated containers managed by atom-runtime on Docker/Kubernetes.
 - **NexusAI - Enterprise LLM Gateway:** *LiteLLM, FastAPI, Python, PostgreSQL, Redis, Docker*
Architected and built NexusAI, an internal multi-tenant LLM gateway built on LiteLLM serving enterprise BFSI clients. Implemented API key management, per-tenant budget governance, PII-aware routing, and audit logging.
- **IHX - A Perfios Company** Bangalore, India
Senior Software Engineer 01-07-2024 - 06-02-2026
 - **Orchestration Engine:** *Netflix Conductor, Zookeeper, Redis Sentinel, TypeScript*
Architected and implemented comprehensive workflow orchestration using Netflix Conductor for document processing pipelines, managing complex extraction, OCR, classification, and quality control tasks. Designed dynamic workflow definitions with parallel processing patterns, reducing processing time by 40% and improving system reliability.
 - **Persistence Engine:** *TypeScript, TypeORM, PostgreSQL, Redis*
Architected and developed a single-layer persistence solution using TypeORM and Node.js/TypeScript that reduced average query time from 1-2 seconds to 10-15ms across 10 tables, significantly improving system performance and scalability.
 - **Status Notification Service:** *TypeScript, AWS SQS, AWS SNS, MongoDB*
Built a high-throughput system processing over 10 million messages daily using SNS to manage and track processes across various AI/ML entities, providing unified status visibility to clients through real-time dashboards.
- **IHX - A Perfios Company** Bangalore, India
Software Engineer 16-08-2022 - 30-06-2024
 - **Document Intelligence Platform:** *TypeScript, FastAPI, AWS EKS, MongoDB, S3*
Developed a comprehensive document processing solution for India's largest health benefits administrator that increased complete claim submissions on first instance from 23% to 77% within 2 weeks. The same platform now classifies over 4 million medical images every month.
 - **Infrastructure Transformation:** *Kubernetes, Helm, NGINX, Grafana, Loki, Promtail, Prometheus, Service Accounts*
Led infrastructure design and implementation for securing AI models and establishing efficient scaling strategies using Kubernetes and AWS services. Introduced comprehensive monitoring with Grafana, Loki, and Prometheus, enabling rapid deployment of new features.
 - **Authentication Solution:** *Keycloak, Java, OAuth 2.0, JWT, NGINX*
Designed and implemented a comprehensive MFA and SSO solution using Keycloak with custom plugins, enabling secure access control to multiple services and applications while reducing login time by 50%.

PROJECTS

- **ATOM - Agentic Development & Deployment Platform (TypeScript, Golang, FastAPI):** Open-source single-tenant agentic platform for running compliant AI agents on-premises. Developers write agents via SDK and deploy with a single command - GATE enforces auth, OPA policy, rate limits, and an immutable audit log on every LLM call. Agents run as isolated containers on Docker/Kubernetes with a Studio UI for lifecycle and HITL management. Adopted and extended at Mphasis for enterprise BFSI deployments.  github.com/shreyasY2k/atom
- **Secure Files Demo (Django, React, Keycloak):** Demo application showcasing MFA plugin integration with self-signed certificates and role-based access control.  github.com/shreyasY2k/secure-files
- **Contra (Three.js, AI-assisted):** Built a complete 3D run & gun shooter game in just 2 hours leveraging AI tools like GitHub Copilot and Claude. Features advanced enemy AI, vehicle physics, and weapon systems with particle effects.  shreyasy2k.github.io/contra
- **Python Spawn (Node.js, npm package):** NPM package bridging Node.js and Python, enabling seamless execution of Python scripts and functions from Node.js applications.  github.com/shreyasY2k/python-spawn

EDUCATION

- **Sapthagiri College of Engineering** Bangalore, India
- *Bachelor of Engineering - Information Science; CGPA: 8.5* 2018 - 2022

TECHNICAL WRITING

- **Building a Custom MFA Plugin for Keycloak:**  SMS, Email & Telegram Integration
- **Mastering Multi-Factor Authentication in Keycloak:**  SMS, Email, and TOTP Setup Guide
- **Unlocking Seamless Access:**  Orchestrating Centralized User Management with Keycloak

CERTIFICATIONS

- **Architecting with Compute Engine:**  Coursera Specialization Certificate
- **The Modern Javascript Bootcamp:**  Udemy Certificate